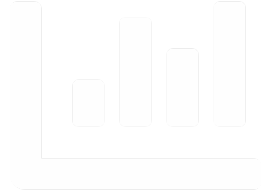


Topic 3



Lecture 3 · Data Collection

National University of Singapore (NUS)

TODAY'S OBJECTIVES

- 1 Lurking and Confounding Variables
- 2 Observational and Experimental Studies
- 3 Good Ways to Sample for Observational Studies
- 4 Experimental Studies



01

Lurking and Confounding Variables

Example 1: Education and Crime

- Dataset collected in different counties in Florida, USA

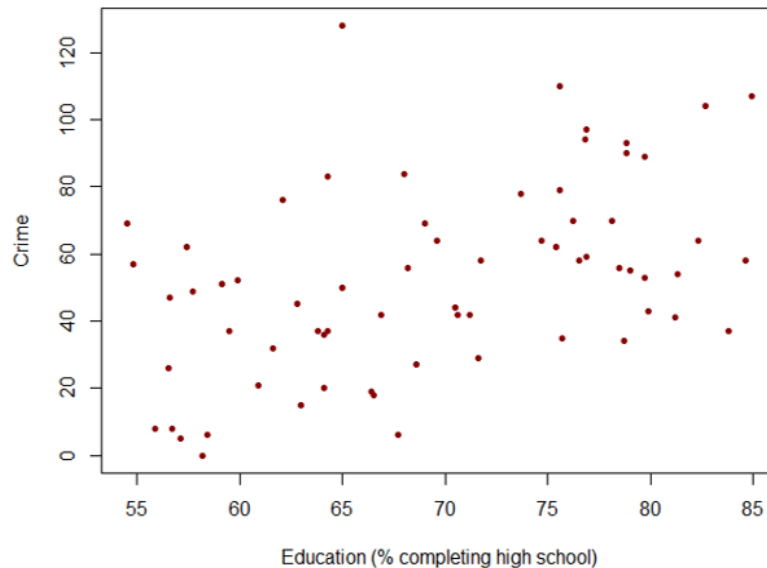
Variable Name	Description	Quantitative / Categorical
Crime (Response)	Number of crimes in that county in the past year	Quantitative
Education Rate (Explanatory)	Percentage of residents, aged at least 25, who completed high school	Quantitative
Urbanization Level	Percentage of residents living in a metropolitan area	Quantitative

Example 1: Education and Crime

- Dataset collected in different counties in Florida, USA
- Aim of study: To investigate the association between education and crime
 - Will education positively or negatively affect the crime rate?
- Both are quantitative variables
- What figure or plot should we use to visualise the association between them?
 - Scatterplot!
- Calculate correlation value

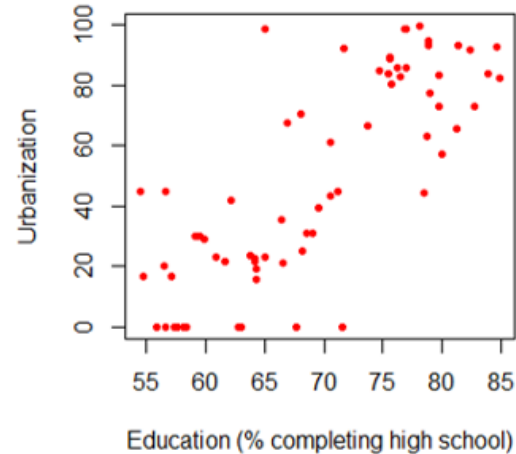
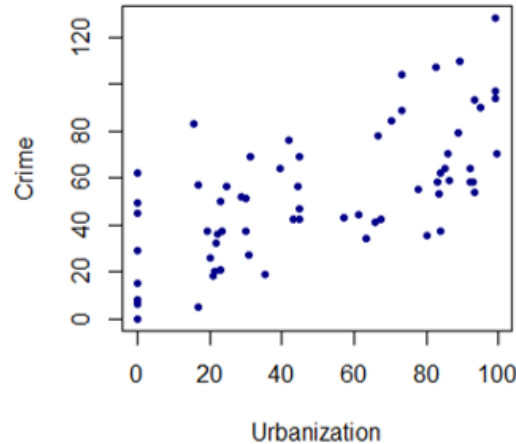
Example 1: Education and Crime

- Association between two variables
 - Might not be very strong
 - Points are not very close to a straight line
 - Positive trend
- $\text{Cor}(X, Y) = 0.47$
 - X : education
 - Y : crime



Example 1: Education and Crime

- Investigate the association of urbanization with education rate and crime
- Positive relationship in both
- Points not very close to straight lines in both



Example 1: Education and Crime

- Crime rate affected positively by urbanization
- Education rate and urbanization are positively correlated
- $\text{Cor}(\text{urbanization}, \text{crime}) = 0.68$
 - $\text{Cor}(\text{crime}, \text{education rate}) = 0.47$
- $\text{Cor}(\text{urbanization}, \text{education rate}) = 0.79$
- Education rate is higher for more highly urbanized counties
- Crime is higher in such counties
 - Chance for crimes to happen increases
 - Might not be because of higher education rate

Example 1: Education and Crime

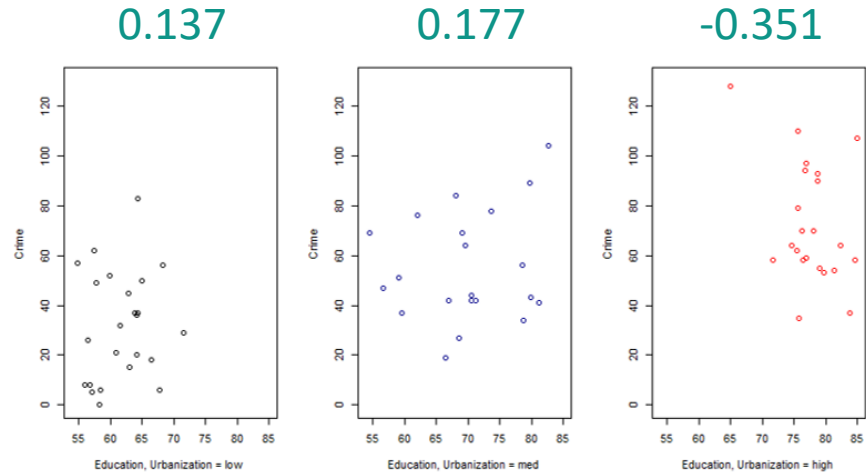
- Condition on urbanization
 - Consider correlation between crime and education rate for different levels of urbanization

Urbanization Level	
0 to 30	Low
30 to 75	Medium
> 75	High

Categorical

Example 1: Education and Crime

- Produce scatterplots of crime and education rate for each category of urbanization
- Low / Medium urbanization: association between crime and education rate is very low
- High urbanization: association between crime and education rate is slightly stronger but negative



Correlation

- Correlation does not imply causation!
- $\text{Cor}(X, Y) = \text{Cor}(Y, X)$
- Cannot conclude that X causes Y or Y causes X

Lurking Variable



DEFINITION

Lurking variable is a variable, usually **unobserved**, that influences the association between the variables of primary interest.

- The whole dataset does not have any record of that variable

Confounding Variable



DEFINITION

Confounding occurs when two explanatory variables are associated with a response variable, but are also associated with each other.

- Observed and recorded in the data
- Difficult to tell which two explanatory variables cause the change in the response variable

Example 1: Education and Crime

LURKING VARIABLE

- Employment rate
 - Not recorded in dataset

CONFOUNDING VARIABLE

- Urbanization
 - Recorded in dataset
 - Associated with education rate and crime

02

Observational and Experimental Studies

Example 2: Cell Phone Usage

- Does cell phone usage have any bad effect on our health?
- First Study:
 - Conducted in Germany 2001
 - Comparing 118 eye-cancer patients with 475 healthy individuals
 - Queried on cell phone usage

Response Variable	Explanatory Variable
Status of having eye-cancer	Frequency of cell phone usage

Example 2: Cell Phone Usage

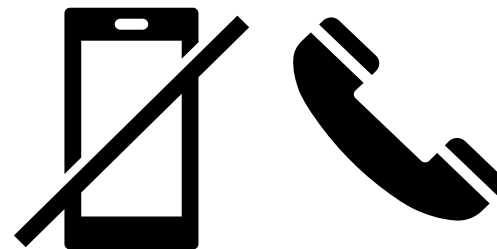
- Does cell phone usage have any bad effect on our health?
- Second Study:
 - Conducted in British 2006
 - Comparing 966 brain-cancer patients with 1716 healthy individuals
 - Queried on cell phone usage

Response Variable	Explanatory Variable
Status of having brain-cancer	Frequency of cell phone usage

Example 2: Cell Phone Usage

- Does cell phone usage have any bad effect on our health?
- Third Study:
 - Conducted in 2011
 - 47 participants with a cellphone attached to their ears twice
 - Comparing change in brain activity monitored by a PET scan
 - Participants do not know if the phone was on or off

Treatment



Response Variable	Explanatory Variable
Measurement of brain activity	Presence / Absence of phone call

Observational Studies



DEFINITION

In observational studies, the values for the response variable and explanatory variable are observed for the sampled subjects or participants, without anything done to them.

EXAMPLE

- First Study
- Second Study

Experimental Study



DEFINITION

An experiment is conducted by assigning subjects to certain experimental conditions, called treatments, and then observing the outcome on the response variable.

Observational VS Experimental Studies

DISADVANTAGES OF OBSERVATIONAL STUDIES

- Lurking variables can affect the results
- First Study:
 - High computer usage could lead to higher chances of eye-cancer
 - Response variable and increase in computer usage could be positively associated with cell phone usage
 - Lurking variable: computer usage

ADVANTAGES OF EXPERIMENTAL STUDIES

- Control lurking variables by randomly assigning treatments
- Third study:
 - Each participant carried the phone twice
 - Whether or not the phone had a call was randomly assigned
 - Participants do not know if the phone had a call or not
- More confidence in determining correlation

Observational VS Experimental Studies

ADVANTAGES OF OBSERVATIONAL STUDIES

- Cheap
- Quick
- Easy to conduct

DISADVANTAGES OF EXPERIMENTAL STUDIES

- Expensive
- Time consuming
- Unethical to put participants at some level of risk

Observational or Experimental?

1

We measure the candy consumption of various countries, along with the population. Our goal is to estimate the candy consumption of Singapore.

2

We wish to investigate the ability of a new drug to reduce hypertension. We recruit a set of participants, and randomly assign either a placebo or the new drug to each of them.

Observational or Experimental?

3

We wish to investigate the use of a post-menopausal hormone drug and its association with breast cancer. We obtain a list of breast cancer patients currently undergoing treatment at NUH and telephone them to ask if they had used this drug in the past 10 years. We also obtain a list of patients at NUH who did not have breast cancer, and telephoned to ask them the same question.

03

Good Ways to Sample for Observational Studies

Sample Survey



DEFINITION

A sample survey is a study that asks questions or takes measurements of the subjects in a sample randomly drawn from the population, in hopes of learning something about the entire population.

Sample Survey

STEP-BY-STEP

1

Identify the population

- "NUS student preferences": all NUS students

2

Compile a list of subjects in the population from which the sample is taken

- List of subjects is known as sampling frame
- "NUS student preferences": all NUS students
- Ideal: sampling frame lists all subjects in population
- Population too large: sampling frame is a subset of population

3

Sampling Design

- Specify a method for selecting subjects from the sampling frame
- "NUS student preferences": determine method to choose a set of 100 students

Convenience

Chance

4

Collect data from chosen sample

- "NUS student preferences": approach them for information needed

Simple Random Sampling



DEFINITION

A simple random sample of n subjects from a population is a sample in which each possible sample of size n has the same chance of being selected.

- Representative of the population
- Allows us to make inferences about the population

Simple Random Sampling

STEP-BY-STEP

1

Number all subjects in the sampling frame

2

Generate a set of n random numbers

- Numbers cannot be repeated
- Numbers are within the range in step 1

3

Select the subjects with the same number as those chosen in step 2

Other ways to obtain random samples

- Cluster Random Sampling
- Stratified Random Sampling

Collecting Data from a Chosen Sample

- Sample of size n chosen in step 3
- Contact them to collect information for data

PERSONAL FACE-TO-FACE INTERVIEW

- Advantage: Easier to get people to respond
- Disadvantage: Costly

TELEPHONE INTERVIEW

- Advantage: Cheaper
- Disadvantage: Easier for people to refuse to participate

SELF-ADMINISTERED QUESTIONNAIRE

- Advantage:
 - Cheaper
 - Less labor-intensive
- Disadvantage: Easier for people to refuse to participate

Sampling Bias

- Sample chosen is not random
- Happens when:
 - Sampling design in step 3 did not use randomization
 - Sampling frame does not represent the full population (under-coverage)
- Example:
 - Aim to study the monthly income of population
 - Sampling frame excludes PRs, residents with EP / long-term visit pass, etc.
 - Under-coverage occurs
- Example:
 - Using telephone survey to collect information
 - Excludes people without a registered phone

Non-Sampling Bias

- Happens in Step 4, collecting data from the sample
- Non-Response Bias:
 - Occurs when some subjects cannot be reached or subjects refuse to participate
 - "NUS student preferences":
 - some students cannot be contacted
 - Uncontactable students possibly study more hours?
 - Results might be biased towards lower study hours per week
- Response Bias:
 - Occurs when participants are dishonest or answer incorrectly
 - In telephone / face-to-face interviews, manner of asking questions could lead subject to answer in a particular way, causing bias

NON-SAMPLING BIAS

```
graph TD; A[NON-SAMPLING BIAS] --- B[SAMPLING BIAS]; A --- C[NON-RESPONSE BIAS]; A --- D[RESPONSE BIAS];
```

SAMPLING BIAS

- Occurs in Step 3, Sampling Design
- Sample is not random
- Under-coverage

NON-RESPONSE BIAS

- Occurs in Step 4, Collecting Data
- Participants cannot be contacted
- Participants refuse to participate

RESPONSE BIAS

- Occurs in Step 4, Collecting Data
- Participants are dishonest or answer incorrectly
- Misleading questions

Poor Alternatives to Simple Random Sampling

- Convenience Samples:
 - Sample selected based on ease of access
- Volunteer Samples:
 - People are encouraged to participate via flyers or email
 - People with strong opinions usually actively join such volunteer surveys
 - May make incorrect inferences when analysing data from volunteer surveys

Example 3: US Presidential Election Poll

- Poll by Literary Digest to predict results of presidential election between Franklin Roosevelt from Democrats and Alf Landon from Republicans
- Sampling frame constructed from telephone directories, country club memberships and automobile registrations
- Questionnaires mailed to 10 million people asking how they planned to vote
- 2.3 million returned the questionnaire, with 57% voting for the Republicans
- In reality, Republicans received only 36% while Democrats won

Example 3: US Presidential Election Poll

- Sampling bias:
 - Due to under-coverage
 - US was in the Great Depression, so those with cars and country club memberships are likely wealthy
 - The wealthy tended to like Republicans
 - Many voters excluded from the sampling frame
 - Subject in sampling frame might not be a registered voter
- Non-response bias:
 - 7.7 million people did not respond
 - Individuals unhappy with Democrats are more likely to respond
 - Thus, the survey predicted the Republicans would win but Democrats won in reality
- A large sample size does not guarantee an unbiased sample

04

Experimental Studies

Experimental Studies

- Subjects or participants of the experiment are referred to as experimental units
- Assign each subject to an experimental condition, called treatment
- Then record the outcome on the response variable
- Goal of an experiment is to investigate the association of how treatment affects response
- Advantage: provides stronger evidence for causation

Elements of a Good Experimental Study

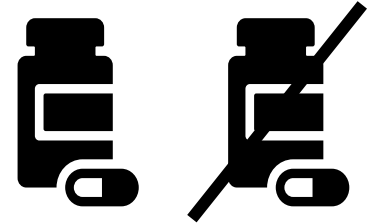
- Control comparison group
- Randomization:
 - Assign treatments to participants randomly
- Blind the study:
 - Subjects do not know if they receive the actual treatment or a placebo

Example 4: Antidepressants for Quitting Smoking

- Purpose: to know if antidepressants help people quit smoking
- Recruited 429 men and women, aged at least 18, who had smoked at least 15 cigarettes a day for the previous year
- Subjects are highly motivated to quit
- Split to two groups:
 - One group took 300mg of antidepressants daily
 - One group did not take any antidepressants
- Observe if subjects quit smoking after one year

Example 4: Antidepressants for Quitting Smoking

- Experimental units: 429 individuals participating in the study
- Descriptive statistics: percentage of people who successfully quit smoking in each group



Response Variable	Explanatory Variable
Results of quitting (successfully quit / failed to quit)	Treatment

Example 4: Antidepressants for Quitting Smoking

- Purpose: to know if antidepressants help people quit smoking
- Recruited 429 men and women, aged at least 18, who had smoked at least 15 cigarettes a day for the previous year
- Subjects are highly motivated to quit
- Split to two groups **randomly**:
 - One group took 300mg of antidepressants daily
 - One group took 300mg of the placebo daily
 - Participants do not know if they were given antidepressants or the placebo
- Observe if subjects 100% quit smoking or still smokes (record number of cigarettes per day) after one year

} Similar size

Randomization

- Use randomization when assigning subjects to treatments
- Eliminates bias that may arise if assigning subjects by hand
- Balances the groups on lurking variables that we know affect the response
- Balances the groups on lurking variables that are unknown to us

Key Takeaways

- 1 Lurking and Confounding Variables
- 2 Observational VS Experimental Studies
- 3 Sample Survey (Simple Random Sampling, Data Collection Methods, Biases)
- 4 Conducting Good Experimental Studies